III B.Tech II Semester (E.C.E)

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23A04602P	**MICROWAVE AND OPTICAL COMMUNICATIONS LAB**	0	0	3	1.5

Course Objectives:

1. To understand the working of microwave bench set up and characteristics of microwave sources.
2. To verify the characteristics of various microwave components and to draw the radiation pattern of antennas.
3. To verify the characteristics of optical sources & detectors and to study about losses in optical fiber.

Course Outcomes:

At the end of this course, the students will be able to

1. Understand the working of microwave bench set up and characteristics of microwave sources.
2. Verify the characteristics of various microwave components and to draw the radiation pattern of antennas.
3. Verify the characteristics of optical sources & detectors and to study about losses in optical fiber.

PART-A: Microwave Lab - Any Seven (7) Experiments

1. Reflex Klystron Characteristics
2. Gunn Diode Characteristics
3. Attenuation Measurement
4. Directional Coupler Characteristics
5. VSWR Measurement
6. Impedance Measurements
7. Frequency and Wavelength measurement
8. Scattering Parameters of Directional coupler
9. Scattering Parameters of Magic TEE
10. Radiation pattern measurement of a Antenna
11. Antenna gain measurement

Part B: Optical Fiber Lab - Any five (5) Experiments

1. Characterization of LED
2. Characterization of Laser Diode
3. Intensity Modulation of Laser output through Optical fiber
4. Measurement of data rate for digital Optical link
5. Measurement of Numerical Aperture.
6. Measurement of Losses for Analog optical link